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Mechanistic studies on Cu(NO₃)₂/TEMPO-catalyzed aerobic oxidation of alcohols to carboxylic acids†Yibo Yu, ^a Zhao Sun, ^b Yin-Long Guo, ^b Xue Zhang, ^b Hui Qian ^a and Shengming Ma ^{a,b}

Oxidation of alcohols to carbonyl compounds is one of the most important and fundamental reactions in organic chemistry. Such aerobic reactions that provide aldehydes or ketones mediated by copper and nitroxyl have been well developed during the past several decades. Recently, we successfully realized the aerobic oxidation of primary alcohols to carboxylic acids catalyzed by Cu/TEMPO for the first time. Herein, a mechanism involving a Cu(NO₃)₂-TEMPO adduct as identified by SAESI-MS studies is proposed, which is supported by our experimental studies and density functional theory (DFT) calculations.

Introduction

Aerobic alcohol oxidation to aldehydes and ketones catalyzed by copper and nitroxyl has been well investigated,^{1,2} and numerous efforts have been made to elucidate the mechanism of these reactions, especially mediated by Cu(II), in recent years.³ In 1984, Semmelhack *et al.* developed the first selective Cu/TEMPO-catalyzed aerobic oxidation of primary alcohols to aldehydes and proposed the engagement of oxoammonium species (Scheme 1A).^{1a} However, Sheldon *et al.* demonstrated that an oxoammonium based mechanism was inconsistent with experimental data, and postulated a mechanism involving an η² Cu-TEMPO adduct, which was analogous to that observed for galactose oxidase mimics (Scheme 1B).^{3a,b} Furthermore, a six-membered transition state involving an η¹ Cu-TEMPO adduct was shown to be favorable with 2,2'-bipyridine (bpy) as a ligand and *N*-methylimidazole (NMI) as a base, wherein Cu(II) and TEMPO served as an one-electron oxidant, respectively (Scheme 1C).^{3d} Recently, we have for the first time reported a Cu/TEMPO-catalyzed aerobic oxidation of alcohols to carboxylic acids under ligand- and base-free conditions.⁴ Here, we wish to report the results of our recent study on the mechanism of this copper nitrate/TEMPO catalyzed aerobic oxidation of alcohols to carboxylic acids.

Results and discussion

Control experiments and determination of the reaction orders

In order to unveil the role of each catalyst, some control experiments for the aerobic oxidation of alcohol **1a** were conducted (Table 1). These results showed that in the absence of Cu(NO₃)₂·3H₂O or TEMPO, the reaction could not occur (entries 2 and 3). When the reaction was carried out in the absence of KHSO₄, only 61% NMR yield of aldehyde **2a'** was obtained with 21% recovery of **1a** (entry 4). The kinetic studies illustrate that it follows the zero-order reaction law from alcohol to aldehyde while first-order from aldehyde to acid (Fig. 1).

Detection of NO_x and verification of the possibility of TEMPO⁺ as the active catalytic species

In addition, we carried out the reaction under standard conditions *via* monitoring with a flue gas analyzer (Scheme 2A, eqn (1)). 12 ppm of NO was detected after 1 h and its concentration was increased to 17 ppm with the formation of 4.1 ppm of NO₂ after 4 h with 19% NMR yield of aldehyde **2a'**. In comparison, for the reaction in the absence of KHSO₄, NO was not observed after 1 h and only 7 ppm of NO was detected after 4 h with 4% NMR yield of aldehyde **2a'** (Scheme 2A, eqn (2)). These results indicated that KHSO₄ could promote the formation of NO_x to facilitate the oxidation reaction.

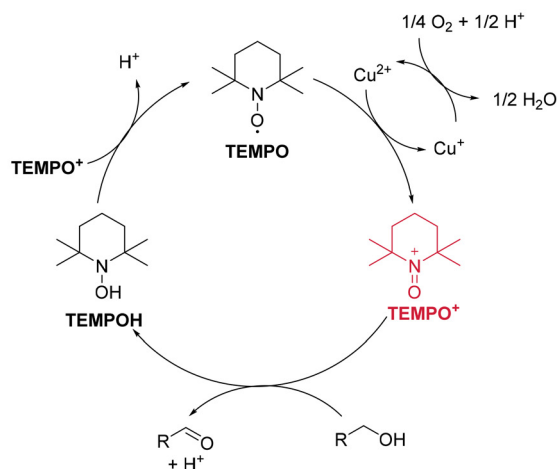
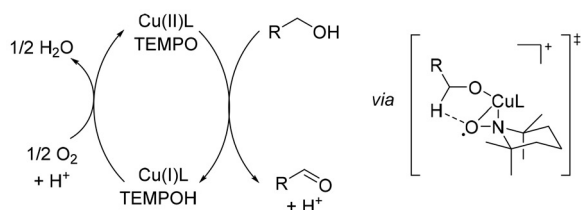
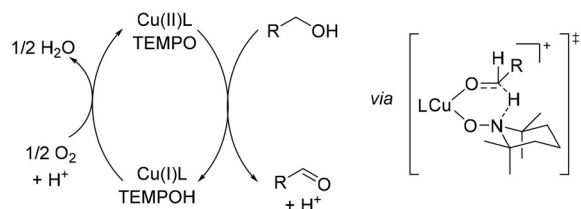
Stahl *et al.*⁵ reported that TEMPO⁺ would be formed from TEMPO in the presence of Cu²⁺ and NO₃⁻ and Anelli and Montanari *et al.*⁶ reported that alcohols could be oxidized to aldehydes by TEMPO⁺ with the formation of TEMPOH, so *N*-oxoammonium salt TEMPO⁺NO₃⁻ was deliberately prepared following the known procedure⁷ and used as the catalyst. It was observed that aldehyde **2a'** was produced in 16% NMR yield with 78% recovery of alcohol **1a** and no carboxylic acid **2a** was formed (Scheme 2B, eqn (3)). The reaction was greatly

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(A) *via* Oxoammonium cations(B) *via* η^2 Cu-TEMPO adduct(C) *via* η^1 Cu-TEMPO adduct

Scheme 1 The reported mechanisms for Cu(II)/TEMPO-catalyzed aerobic alcohol oxidation to aldehydes.

accelerated when 10 mol% of NO_2 was used together with 10 mol% of $\text{TEMPO}^+\text{NO}_3^-$; however, still aldehyde **2a'** was formed as the major product in 69% yield with only 21% yield

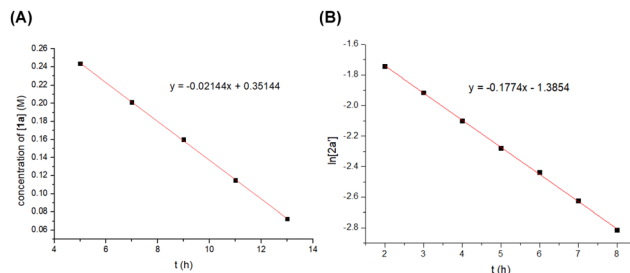


Fig. 1 (A) Determination of the order for alcohol **1a**. (B) Determination of the order for aldehyde **2a'**.

of carboxylic acid **2a** (Scheme 2B, eqn (4)). These results illustrated that TEMPO^+ was not the major active catalytic species. In addition, when 10 mol% of TEMPO and 20 mol% of NO_2 were applied for the aerobic oxidation of aldehyde **2a'**, 29% of acid **2a** was obtained with 59% recovery of **2a'** (Scheme 2C, eqn (5)). In comparison, aldehyde **2a'** was smoothly oxidized to carboxylic acid **2a** in 80% yield under the standard conditions (Scheme 2C, eqn (6)). These results suggested that the oxidation of aldehyde *via* a copper-free NO_x/TEMPO catalytic cycle⁸ was at least not the major pathway although it could not be ruled out completely.

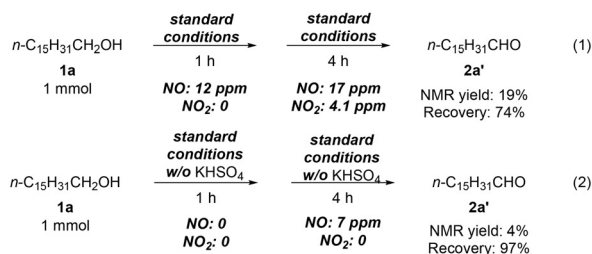
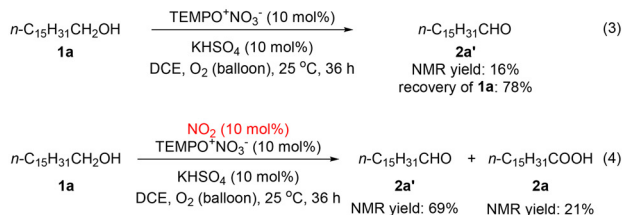
¹⁸O labelling experiments

Furthermore, ¹⁸O labelling experiments with alcohol **1a** as the substrate were conducted (Scheme 3A). When H_2^{18}O and O_2 were added to the reaction, 20% of single ¹⁸O-labeled carboxylic acid **2a** was observed (Scheme 3A, eqn (1)). However, 42% of single ¹⁸O-labeled carboxylic acid and 8% of double ¹⁸O-labeled carboxylic acid **2a** were obtained when $^{18}\text{O}_2$ were applied (Scheme 3A, eqn (2)). Then, ¹⁸O labelling experiments with aldehyde **2a'** as the substrate were conducted (Scheme 3B): 33% of single ¹⁸O-labeled carboxylic acid and 6% of double ¹⁸O-labeled carboxylic acid were observed when $^{18}\text{O}_2$ was used (Scheme 3B, eqn (3)) while 37% of single ¹⁸O-labeled carboxylic acid and 7% of double ¹⁸O-labeled carboxylic acid were obtained when O_2 and 1.0 equivalent of H_2^{18}O were applied (Scheme 3B, eqn (4)). Considering that $^{18}\text{O}_2$ was in

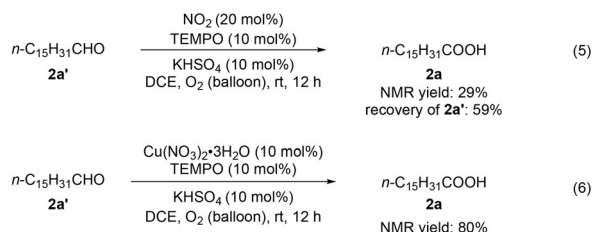
Table 1 Control experiments for the aerobic oxidation of alcohol **1a**^a

$ \begin{array}{c} \text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O} \text{ (10 mol\%)} \\ \text{TEMPO} \text{ (10 mol\%)} \\ \text{KHSO}_4 \text{ (10 mol\%)} \\ \text{DCE, O}_2 \text{ (balloon), rt, 36 h} \\ \text{"standard conditions"} \end{array} $						
Entry	$\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	TEMPO	KHSO ₄	Recovery of 1a (%)	NMR yield of 2a' (%)	NMR yield of 2a (%)
1	+	+	+	0	0	100
2	—	+	+	~100	0	0
3	+	—	+	~100	0	0
4	+	+	—	21	61	0

^a The reaction was conducted with 1.0 mmol of **1a** in 4 mL of DCE at rt for 36 h with an O_2 balloon. The NMR yield and recovery were determined by ¹H NMR analysis using dibromomethane as the internal standard.

(A) The effect of KHSO₄(B) TEMPO⁺NO₃⁻ as catalyst

(C) Oxidation of aldehyde 2a' to carboxylic acid 2a

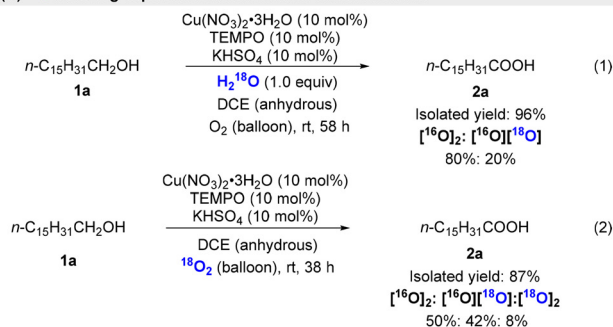
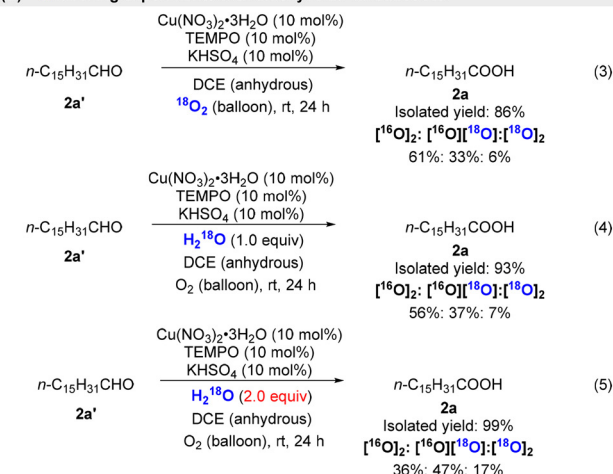


Scheme 2 Detection of NO_x and verification of the possibility of TEMPO⁺ as the active catalytic species.

excess, the amount of H₂¹⁸O was increased to 2.0 equiv., and we found that the yield of the ¹⁸O-labeled product was significantly improved (Scheme 3B, eqn (5)). These results indicated that the main source of the second oxygen atom is ambient H₂O with the hydrated aldehyde being the possible intermediate. The formation of double ¹⁸O-labeled carboxylic acid also demonstrated a reversible pathway between aldehyde and aldehyde hydrate.

Identification of the Cu(NO₃)₂-TEMPO adduct

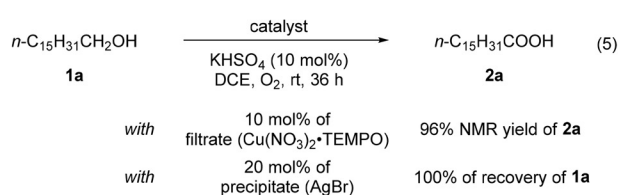
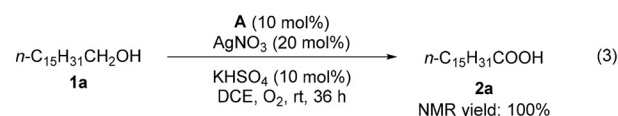
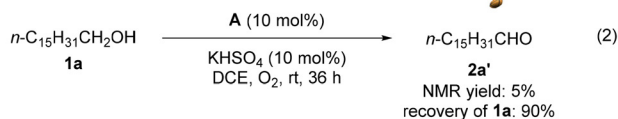
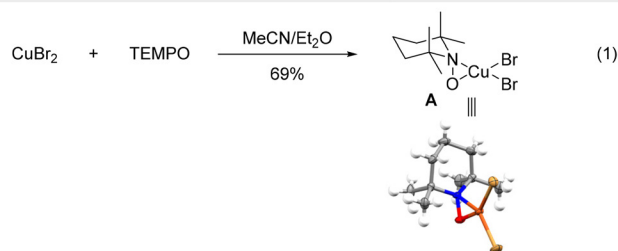
In order to further identify any possible catalytically active species, the reaction of Cu(NO₃)₂·3H₂O and TEMPO under different conditions was tried; however, no crystals were obtained. Fortunately, the reaction of CuBr₂ with TEMPO afforded complex A,⁹ whose structure has been successfully established by X-ray diffraction study (Fig. 2A, eqn (1)). 5% NMR yield of aldehyde 2a' was obtained with 90% recovery of alcohol 1a when the reaction was conducted with 10 mol% of complex A and KHSO₄ (Fig. 2A, eqn (2)). Interestingly, when AgNO₃ (20 mol%) was added to exchange the Br⁻ out of complex A with NO₃⁻, a quantitative yield of carboxylic acid 2a was obtained from alcohol 1a (Fig. 2A, eqn (3)). To exclude the effect of AgBr, we carried out the reaction of complex A and

(A) ¹⁸O labelling experiments with alcohol 1a as substrate(B) ¹⁸O labelling experiments with aldehyde 2a' as substrate

Scheme 3 ¹⁸O labelling experiments.

AgNO₃ (Fig. 2A, eqn (4)). After filtration, the filtrate and precipitate were used as the catalyst, respectively. Carboxylic acid 2a was formed in 96% yield with the filtrate as the catalyst while no reaction occurred with the precipitate (AgBr) as the catalyst (Fig. 2A, eqn (5)). Solvent-assisted electrospray ionization mass spectrometry (SAESI-MS) studies¹⁰ were conducted to analyze any possible species from the diluted filtrate. Signals with *m/z* of 281 and 283 were observed, which are consistent with the *m/z* of [TEMPO-Cu(NO₃)]⁺ (Fig. 2B-i, calcd for C₉H₁₈⁶³CuN₂O₄⁺: 281; C₉H₁₈⁶⁵CuN₂O₄⁺: 283). For further structural characterization of this intermediate, a SAESI-MS/MS experiment was conducted: the corresponding signal of TEMPO⁺ was detected (Fig. 2B-ii, calcd for C₉H₁₈NO⁺: 156), formed by losing Cu²⁺, NO₃⁻ and an electron. These results illustrate the existence of the Cu(NO₃)₂-TEMPO adduct and support that A-1-type species may be the catalytically active species (Scheme 4).

Based on these mechanistic studies, a possible mechanism has been proposed (Scheme 4): the η² Cu-TEMPO species A-1 could undergo a ligand exchange reaction with the alcohol to form A-2. TEMPOH, aldehyde 2', and Cu⁺ would be generated *via* H-abstraction from a six-membered cyclic transition state TS-1. TEMPO would be regenerated *via* the oxidation of TEMPOH by Cu²⁺,^{3e} which would be recycled by the oxidation of Cu⁺ with NO₂. Subsequently, water would attack the acti-

(A) Aerobic oxidation of alcohol **1a** with **A** as catalyst

(B) SAESI-MS studies

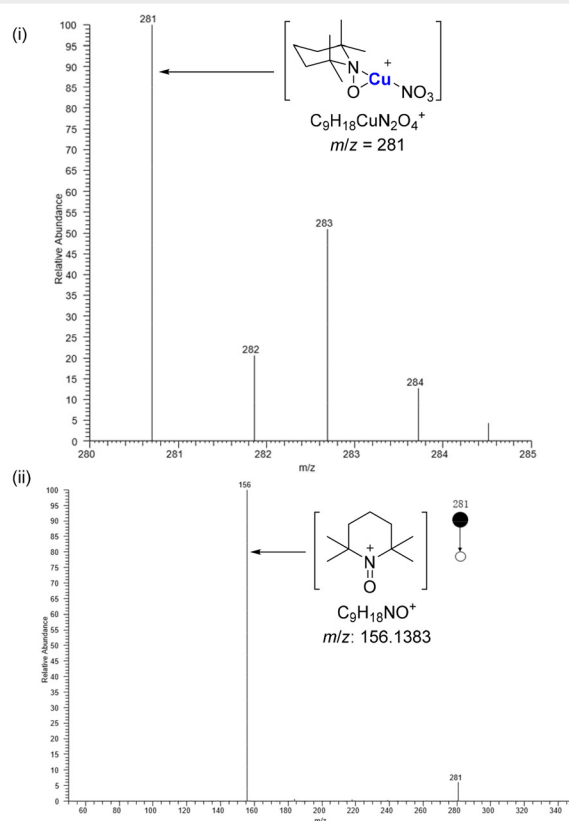
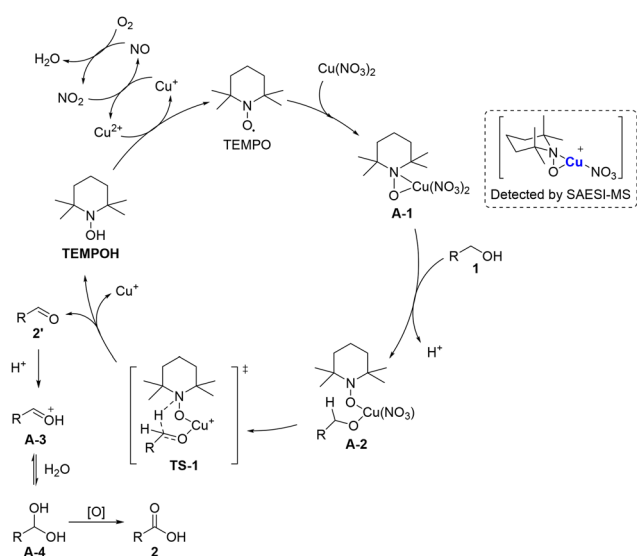


Fig. 2 Identification of the $\text{Cu}(\text{NO}_3)_2$ -TEMPO adduct. (A) Aerobic oxidation of alcohol **1a** with **A** as the catalyst. (B) SAESI-MS studies. (i) SAESI-MS spectrum of the diluted filtrate showing the signal of $[\text{TEMPO-Cu}(\text{NO}_3)]^+$ at m/z 281. (ii) SAESI-MS/MS spectrum of the complex ion $[\text{TEMPO-Cu}(\text{NO}_3)]^+$ at m/z 281.



Scheme 4 The proposed mechanism.

vated aldehyde intermediate **A-3** to form the aldehyde hydrate **A-4**, which was further oxidized to carboxylic acid through the same process as the oxidation of alcohol.

DFT calculations

Density functional theory (DFT) calculations were performed at the M06/6-311+G(d,p)-SDD/SMD(DCE)//M06/6-31G(d)-LANL2DZ level of theory to further support this proposed mechanism, using ethanol as the model substrate.¹¹ Initial coupling of TEMPO with $\text{Cu}(\text{NO}_3)_2$ provides the **A-1** type η^2 -TEMPO adduct **int0**, which has a closed-shell singlet electronic structure. Subsequent isomerization of **int0** yields a more stable η^1 -TEMPO adduct **int1**, which coordinates with alcohol to generate intermediate **int2**. Then displacement of one of the nitrate ions by the alkoxide ligand occurs uphill energetically (+21.3 kcal mol⁻¹, **int4** with respect to **int1**) to produce the **A-2** type Cu-alkoxide species **int4**, accompanied by the release of nitric acid. Subsequently, a six-membered-ring transition structure **TS1** is located for the migration of hydrogen from the alkoxide to the nitrogen atom of TEMPO and the simultaneous reduction to Cu(I), resulting in the formation of aldehyde, TEMPOH and CuNO_3 . Notably, the activation barrier for the hydrogen shift is very low (almost no barrier) from **int4**, while the overall free energy for oxidation of alcohol to aldehyde is 21.3 kcal mol⁻¹ (Fig. 3A).

Subsequently, the aldehyde hydrate **int7** could be generated from the reaction of aldehyde with water *via* a four-membered

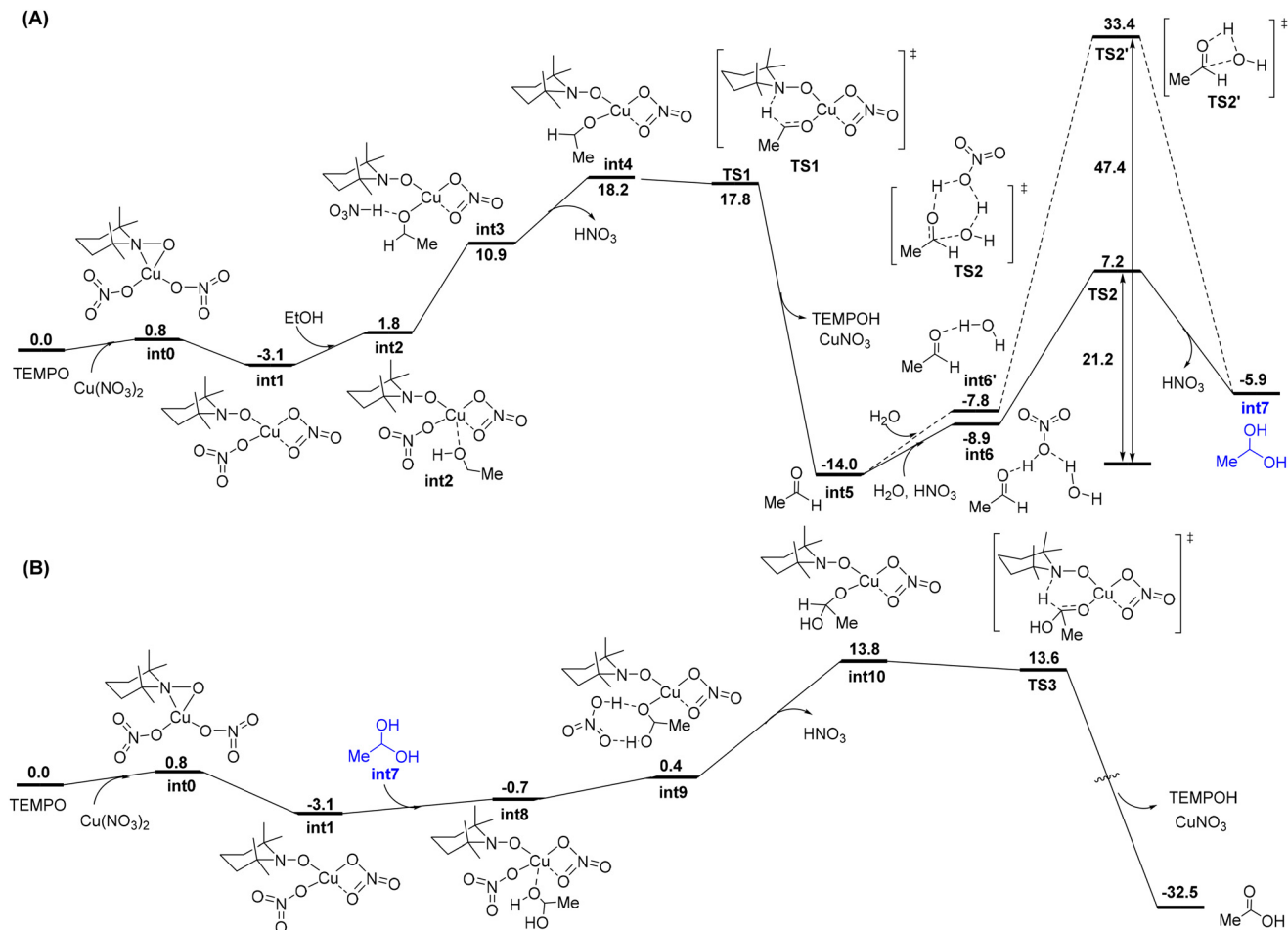


Fig. 3 DFT study of the mechanism shown in Scheme 4. (A) Free energy profiles (ΔG_{sol} in kcal mol⁻¹) for the copper-catalyzed oxidation of alcohol to aldehyde. (B) Free energy profiles (ΔG_{sol} in kcal mol⁻¹) for the copper-catalyzed oxidation of aldehyde hydrate to acetic acid.

cyclic transition structure **TS2'**, which features a very high barrier of 47.4 kcal mol⁻¹. With the aid of HNO₃ as the proton shuttle, this process would take place through a six-membered cyclic transition structure **TS2** with a much lower activation barrier of 21.2 kcal mol⁻¹. Then, aldehyde hydrate **int7** would be further oxidized to acetic acid through a similar process with an overall free energy of 16.9 kcal mol⁻¹ (**int10** with respect to **int1**, Fig. 3B). Based on the mechanistic experiments and DFT calculations, we considered that the active catalytic species is more inclined to be a Cu(III) species (see Scheme S2 in the ESI† for details).

Conclusions

In summary, we present the mechanistic investigations of Cu(NO₃)₂·3H₂O/TEMPO/KHSO₄-catalyzed aerobic oxidation of primary alcohols to carboxylic acids. KHSO₄ is proven to promote the generation of NO_x to facilitate the aerobic alcohol oxidation. On the basis of experimental results, we conclude that TEMPO⁺ is at least not the major active catalytic species.

Furthermore, we observed a Cu(NO₃)₂-TEMPO adduct of **A-1** as supported by SAESI-MS studies, which catalyzed this aerobic alcohol oxidation. DFT calculations have been applied to validate that the Cu(NO₃)₂-TEMPO adduct-catalyzed mechanism is favored over TEMPO⁺ catalysis.

Author contributions

S. Ma, H. Qian and Y. Yu designed the experiments. Y. Yu performed the experiments. Z. Sun and Y.-L. Guo performed the SAESI-MS studies. X. Zhang performed the DFT calculations. Y. Yu, X. Zhang, H. Qian and S. Ma wrote the manuscript.

Data availability

The data supporting this article have been included as part of the ESI.† Crystallographic data for complex **A** have been deposited at the Cambridge Crystallographic Data Center [CCDC 2353933].†

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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- 11 See the ESI† for DFT studies on two other less favorable pathways.